CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The radii of two circles are 5 cm and 12 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 7 cm **(b)** 15 cm **(c)** 17 cm **(d)** 13 cm

2. The minute hand of a clock is 21 cm long. The area swept by the minute hand in 10 minutes is: 1

(a) 22 cm² **(b)** 1386 cm² **(c)** 462 cm² **(d)** 231 cm²

3. A horse is tied at a corner of a large square field with a rope 14 m long. The area of the field it can graze is: 1

(a) 154 m² **(b)** 22 m² **(c)** 616 m² **(d)** 308 m²

4. The area of a sector of a circle of radius 21 cm is 346.5 cm^2. The central angle of the sector is: 1

(a) 120° **(b)** 180° **(c)** 90° **(d)** 45°

5. A circular field has radius 35 m. Fencing costs ₹ 4 per metre. The cost (in ₹) of fencing the field once is: 1

(a) ₹ 220 **(b)** ₹ 440 **(c)** ₹ 15400 **(d)** ₹ 880

6. A wheel has radius 14 cm. The number of complete revolutions it makes to cover 44 m is: 1

(a) 100 **(b)** 25 **(c)** 60 **(d)** 50

7. The radii of two circles are 12 cm and 5 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 34 cm **(b)** 17 cm **(c)** 13 cm **(d)** 8.5 cm

8. The radii of two circles are 9 cm and 12 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 3 cm **(b)** 15 cm **(c)** 21 cm **(d)** 17 cm

9. The minute hand of a clock is 21 cm long. The area swept by the minute hand in 15 minutes is: 1

(a) 1386 cm² **(b)** 346.5 cm² **(c)** 693 cm² **(d)** 33 cm²

10. A horse is tied at a corner of a large square field with a rope 21 m long. The area of the field it can graze is: 1

(a) 346.5 m² **(b)** 33 m² **(c)** 1386 m² **(d)** 693 m²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A wire in the form of a circle of radius 28 cm is bent into a square. Find the area of the square. **2**

12. A horse is tied at a corner of a large square field. The rope is first 14 m long and then lengthened to 21 m. Find the increase in the area it can graze. **2**

13. The minute hand of a clock is 14 cm long. Find the distance moved by its tip in 45 minutes. **2**

14. A wire in the form of a circle of radius 35 cm is bent into a square. Find the area of the square. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A chord of a circle of radius 21 cm subtends a right angle at the centre. Find the area of the minor segment. **3**

16. A sector of a circle of radius 14 cm has central angle 45°. Find the perimeter of the sector. **3**

17. An umbrella has 10 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 35 cm. Find the area between two consecutive ribs. **3**

OR

A chord of a circle of radius 28 cm subtends a right angle at the centre. Find the area of the minor segment. **3**

18. A sector of a circle of radius 14 cm has central angle 270°. Find the perimeter of the sector. **3**

OR

An umbrella has 12 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 42 cm. Find the area between two consecutive ribs. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A chord of a circle of radius 28 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. **5**

OR

A brooch is a circular piece of silver wire of diameter 42 mm. The wire is also used for 6 equal diameters that divide the brooch into 12 sectors. Find the total length of wire required (in mm). **5**

20. A chord of a circle of radius 14 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. **5**

OR

A square is inscribed in a circle of radius 7 cm. Find the area of the circle outside the square. **5**

END OF QUESTION PAPER — **Total: 40 Marks**